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Soils as extended composite phenotypes

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ABSTRACT

Several recent theories and conceptual frameworks in pedology, ecology, geomorphology, and evolutionary biology, taken together, suggest the notion that earth's soils are not just strongly influenced by biota, but represent selective pressures. These ideas point in the direction of many aspects of soils as expressions of the effects of genes through the effects of organisms (i.e., extended phenotypes). The cumulative, interacting, overlapping effects of these extended phenotypes as manifested in the soil represent an *extended composite phenotype*. If this is the case, then we should expect major changes in biological evolution to be reflected in major changes in the types of soils. A brief review of paleopedology literature supports the notion of coevolution of soils and biota, not just in the sense of both responding to the same environmental forcings, but also with respect to explicit pedological expressions of biological change. The extended composite phenotype notion also suggests that significant biological changes should be reflected in significant qualitative pedological changes (i.e., fundamental changes in soil morphology beyond quantitative changes in specific short-lived soil properties). This is demonstrated via recent research in the Ouachita Mountains, Arkansas, where changes in selective pressures on trees associated with fire frequency are linked to changes in soil morphology. The concept of soils as extended composite phenotypes has repercussions for pedology, evolutionary biology, and interpretations of soils in other earth and environmental sciences. The notion of genetic signatures in soil morphology also has implications for the search for extraterrestrial life, and extends the notion of Earth as a set of tightly-coupled, densely interwoven systems.

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1. Introduction

This paper argues that Earth's soils are an expression of the cumulative impacts of the biosphere on surface processes. As the product of the combined, interacting influences of (at least) geology, climate, biota, topography, and time, soils may represent the single best indication of the nature, state, and condition of earth surface systems (Albrecht, 1957; Lovelock, 1993; Retallack, 1994, 2001; Huggett, 1998; Richter and Markewitz, 1995; Phillips, 1999). Thus, if soils are expressions of biological evolution, rather than, or in addition to, being simply influenced by biotic processes, implications are profound for understanding the evolution of earth systems.

"Evolution" is sometimes used in a general sense to refer to change over time, and in some cases explicitly refers to change characterized by contingency and path dependence, as opposed to monotonic, deterministic development. In these senses there is already a wide appreciation of the coevolution of soils, biota, ecosystems, and landforms. If soils are indeed a phenotype of biota, then coevolution of soils and biota also occurs in the specific sense of biological evolution by means of natural selection.

There already exist a number of valid, useful, and complementary ways of viewing soils—as media for plant growth, habitats and ecosystems, weathered mantles, potentially mobile surface debris covers, membranes, reactors, and engineering materials, to name a few (see, e.g., Ollier and Pain, 1996; Schaetzl and Anderson, 2006). Certainly, the role of soils as habitat, growth media, and edaphic factors is widely appreciated, as are the important influences of organisms on soil. Corenblit et al. (2008) included soils and biomantles as one of four examples of ecological engineering over geological timescales, whereby soils (and other geomorphic features) "may be considered as a by-product of the natural selection process of living organisms," which is consistent with the argument in this paper. A view of soils as extended composite phenotypes, which goes well beyond the influences of soils and organisms on each other, and arises from several existing conceptual frameworks, which taken together point to a view of soil as a manifestation of both biotic effects and biological evolution. These include:

- The idea that (many) soils include, or consist of, a *biomantle*, a surficial layer not only containing organisms and influenced by biochemical effects, but continually mixed by floral and faunalturbation (Johnson, 1990; Paton et al., 1995; Johnson, 2002).
- The notion of the *extended phenotype* (Dawkins, 1982), which refers to all effects of a gene on the environment.
- The concepts of *ecosystem engineers* and *niche construction*; which recognize that organisms may not only have profound effects in modifying the environment, but that those environmental

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modifications influence natural selection and evolution (e.g. Jones et al., 1994, 1997; Odling-Smee et al., 2003).

- Emergence of *biogeomorphology* and ecogeomorphology, which includes the notions of tight coupling among biological, ecological, geological, and pedological processes; and that biotic effects may be comparably (or more) influential than geophysical phenomena on some surface processes and landforms (e.g. Stallins, 2006; Corenblit et al., 2007, 2008).
- Vernadsky's (1926) concept of the *biosphere* as a planetary membrane for capturing, storing, and transforming solar energy.

These ideas point in the direction of many aspects of soils as expressions of the effects of genes through the effects of organisms (i.e., extended phenotypes). The cumulative, interacting, overlapping effects of individual extended phenotypes as manifest in the soil represent an *extended composite phenotype*. This notion includes the effects of multiple generations of multiple organisms and may incorporate both positive and negative niche constructions with respect to individuals or species. Thus the concept differs from a strict interpretation of Dawkins (1982) extended phenotype concept, but is consistent with the basic idea of extended phenotypes as environmental expressions of genes which in turn influence natural selection.

2. Conceptual framework

2.1. Soils as biotic constructs

A view of soils as "biotic constructs" was laid out by van Breemen (1993), who argued that soils are influenced by biota in such a way as to increase net primary productivity. van Breemen (1993) reviewed the effects of organisms on soil texture, structure, erosion, organic matter content, chemistry, and outlined biological processes which generally make Earth's crust more favorable for plant growth. The latter include weathering and other physical transformations of rock and sediment, increasing nutrient supply, reducing nutrient losses, increasing water-holding capacity, decreasing compaction, and increasing aeration. He also gave examples of biotic effects on soils which give particular species advantages by stressing or inhibiting competitors.

Mechanisms making a (soil) environment more favorable at an ecosystem scale can develop only if the number of individuals and species contributing to these mechanisms increase over time. The selective mechanisms involved in such an increase are, van Breemen (1993: 205) admitted, "not immediately apparent." Nonetheless, van Breemen argued (via examples) that the spread of traits both advantageous to individuals and having a long-term ecologically favorable effect on the soil can be explained by classical natural selection. He also noted, however, that individually advantageous traits could just as commonly have long-term deleterious effects on soil/ecosystem productivity. The arguments were extended by van Breemen and Finzi (1998), who concluded that there is strong evidence for plant–soil feedbacks in many ecosystems, and that these feedbacks may have played a role in the evolution of the plants involved.

While van Breemen's (1993) discussion paper does not explicitly address phenotypes, it does paint a clear picture of soils as biotic features, and is consistent with the extended composite phenotype concept. Discussions and critiques of the paper are given by Chapin (1993), Lovelock (1993), and Wilding and Kelly (1993). Chapin (1993) specifically addresses selection and evolution with respect to soil development, arguing that biotic effects on soils tend to create more favorable conditions for plants in general during early stages of primary succession, but otherwise are as likely to be negative as positive with respect to productivity.

2.2. Biomantles

Soils, once conceived as a primarily geological mantle and a medium for biota, are increasingly viewed by pedologists as continuously-mixed biomantles (Johnson, 1990; Paton et al., 1995; Balek, 2002; Johnson, 2002; Gabet et al., 2003). This perspective can be traced at least to the late 19th century work of Charles Darwin and Nathaniel Shaler, but only recently have conceptual models of soil and regolith incorporating the idea of continuous bioturbation become widely accepted (Johnson, 2002; Gabet et al., 2003; Meysman et al., 2006). Even in Antarctica, a study of the relationships among vegetation, lithology, climate, and soil properties along a latitudinal gradient in Victoria Land found that vegetation was related to chemistry of the surface soil layer, and to the depth of the active layer. More surprisingly, the study found that penguins had the greatest influence on soil properties, initiating the development of ornithogenic soils (Cannone et al., 2008).

Studies of the considerable biological effects on soils have traditionally concentrated on biochemical, ecological, and edaphic influences. However, biomechanical effects have increasingly been found to be of comparable importance (e.g., Johnson, 1993, 2002; Balek, 2002; Gabet et al., 2003; Phillips and Marion, 2006).

One example is the formation of vertical texture contrast soils (TCS), a globally common phenomenon characterized by coarser surficial layers overlying finer subsoils. While the possibility of multiple modes of formation has long been recognized, traditional pedologic theory has emphasized eluviation–illuviation whereby finer material is removed from upper and deposited in lower layers by percolating water. The dominance of this perspective is evident in the definitions of A, E, B, and C horizons in the most common soil taxonomies (Tandarich et al., 2002).

Sometimes, however, processes other than eluviation–illuviation may be of comparable or greater importance in creating texture contrasts. Paton et al. (1995) presented evidence that illuviation is inadequate as a general global explanation and proposed bioturbation and winnowing as a common cause of textural differentiation. Donald L. Johnson established the role of bioturbation in creating surficial biomantles, and that biomechanical activities can result in the differentiation rather than the homogenization of soil layers (Johnson, 1985; Johnson and Watson-Stegner, 1987; Johnson, 1990; Johnson et al., 2005). Later, he more specifically linked formation of biomantles to texture contrast profiles, as well as incorporating bioturbation and biomantles into general theories of soil and landscape evolution (Johnson, 1993; Johnson and Balek, 1991; Johnson, 1994). Phillips (2004, 2007) reviewed the literature on TCS, and presented a model based on polygenesis and multiple causality, including the formation of biomantles.

2.3. Extended phenotypes and niche construction

Dawkins (1982) argued that restricting the concept of the phenotype to the genetic expression evident in an organism's body is too restrictive, and introduced the idea of the extended phenotype. The extended phenotype includes all the effects of a gene or a combination of genes on the environment, through an organism's actions or behaviors. The distinction between extended phenotype and environmental effects more generally is that the former should benefit the selection of the engineer organism's genes. More obvious examples include features such as ant and termite mounds and beaver dams. Thus organisms and their biotopes (which presumably include soils and aspects thereof) may reciprocally define and construct each other (Dawkins, 1982).

Ecosystem engineers are organisms which have a disproportionate influence on the environment relative to their numbers (Jones et al., 1994, 1997). Ecosystem engineering is the ecological counterpart of the niche construction concept in evolutionary biology. As outlined by Odling-Smee et al. (2003), niche construction is the process by which ecological engineer organisms alter the environment (for themselves and/or other species) so as to influence selective pressures. In contrast to traditional evolutionary theory, focused on organismal adaptations

to the environment, niche construction includes the modification of the environment by organisms.

An extensive literature exists on niche construction, ecosystem engineers, and extended phenotypes (Hastings et al., 1993; Brodie, 2005; Griffiths, 2005). Overviews and discussions of these as they relate to biogeomorphology (Stallins, 2006; Corenblit et al., 2007, 2008) are equally relevant to pedology.

Ecosystem engineering artifacts relevant to natural selection are equivalent to extended phenotypes, and are sometimes termed "true" or positive niche construction. However, environmental alterations may also influence selective pressure on other organisms, and may not favor the genes of the engineer. Thus, while cases of positive niche construction where environmental alterations benefit the engineer organisms may be difficult to unequivocally identify in nature, niche construction and niche changing more generally–environmental modifications which have some form of selective repercussions–is common (Stallins, 2006; Corenblit et al., 2008).

A model with similar implications with respect to two-way adaptations of biota and the abiotic environment was developed by Smith (1986) from energetics principles. The biotic/abiotic sector model is based on the evolution of both biota and environment. While Smith (1986) views the biotic sector as a whole evolving in response to constraints set by the abiotic sector, the necessity of energy/negentropy import requires suitably adapted organisms at the right place and time. Evolution is seen as a function of biotic contributions to development of more efficient resource turnover. Recursive change (responses to and influences on the abiotic environment) within the biotic sector "is most fundamentally guided by selection producing more efficient means of turning over material resources (spatial interaction), as opposed to 'fit' individuals and population (250)." Smith's model (further elaborated in Smith, 1989, 2005) provides direct links between ecological notions of ecosystem engineering, evolutionary biological concepts of niche construction, and biogeographical principles related to changes in ranges and distributions.

Ecosystem engineering and/or niche construction concepts have been explicitly applied to soils in a few cases. One of the primary examples of niche construction presented by Odling-Smee et al. (2003: 11–12, 291, 374–376) is that of earthworms, which are anatomically poorly adapted to life on land. However, the worms' modifications of the soil modify the environment to meet their needs.

Jouquet et al. (2006) reviewed research on soil invertebrates as ecosystem engineers, particularly termites, ants, and earthworms, whose effects on soil properties influence the availability of resources for other organisms. They suggest that strategies evolved by species or functional groups to control their environments can explain the difference between ecosystem engineering or positive niche construction as extended phenotypes and "accidental engineers." The former primarily build biogenic structures to maintain optimal growth conditions for themselves. Accidental engineers, by contrast, expend energy in moving through the soil to be as close as possible to their optimal environment. Extended phenotype engineers tend to maintain ecosystem heterogeneity, according to Jouquet et al. (2006), because they concentrate their activities at a few points, whereas accidental engineers move through the soil and may homogenize nutrient distribution.

In this paper no attempt is made to disentangle effects of extended phenotype and accidental engineers. Rather, the cumulative, interacting effects of organisms are considered as an extended (composite) phenotype.

2.4. Biogeomorphology

The study of interrelationships between landforms and geomorphic processes on one hand and biotic and ecological processes on the other has undergone a transition in recent years. Ecologists have long recognized that erosion, sedimentation, weathering, etc. have important influences on organisms and communities, and that landforms are an important component (and in some cases perhaps the single best indicator) of abiotic habitat characteristics. Likewise, earth scientists have long recognized the important effects of biota on geomorphic processes—e.g., the role of microbes in rock weathering or the effects of vegetation cover on erosion.

These perspectives have recently been expanded into a view recognizing not just effects of geomorphic and biotic factors on each other, but also feedback relationships, tight coupling, and coevolution. This approach is typically called biogeomorphology or ecogeomorphology, which recognizes that surface processes and landforms cannot be fully understood without accounting for the feedbacks and mutual adjustments between biological and geophysical phenomena. Even more recently, the notion of aspects of geomorphology as outcomes of biological evolutionary dynamics has been explicitly incorporated in biogeomorphology (Stallins, 2006; Corenblit et al., 2007, 2008). The evolutionary biology concept of niche construction, Corenblit et al. (2007) noted, provides an opportunity to extend the debate about evolutionary consequences of biotic–abiotic feeedbacks to geomorphology.

2.5. Biosphere and Soil

V.I. Vernadsky, a pioneer of biogeochemistry, saw the biosphere as a web of biogeochemical feedback mechanisms connecting organisms and environment in a single system. In his view, life has a natural tendency for expansion through intensification of biogeochemical cycles, and natural selection favors individuals and species capable of increasing cycling of elements (Vernadsky, 1926). Lapenis (2002), drawing on Vernadsky and several other Russians writing in the late 19th and early 20th century, presented an updated version of these ideas based on a principle of biogeochemical selection. Natural selection, operating at the genetic or individual level, favors better recycling skills. The cumulative impacts of this lead to a tendency to evolve toward a state of greater productivity, diversity, and sociability (Lapenis, 2002).

Particularly relevant in the context of this paper are the ideas of Rizpolozhensky (1892, 1897), who highlighted mutualism as a driving force in evolution—a radical view at the time. According to Rizpolozhensky (via Lapenis, 2002), mutual aid arises for the purpose of joint preparation of food. Organisms change the environment through the formation of "soil" and organisms change the environment ("soil") in a direction favorable for life in general. Rizpolozhensky's notion of soil is apparently broader than contemporary definitions, but may be interpreted as an early exposition of a notion of soils as an expression of the combined, interacting effects of natural selection.

Vernadsky's (1926) view of the biosphere as a planetary membrane for capturing and converting solar energy is broadly analogous to pedologist Nikiforoff's (1959) characterization of soil as "the excited skin" of the subaerial part of Earth's crust. Taken together, Nikiforoff and Vernadsky's conceptual frameworks suggest the notion of soil as a physical manifestation of the biosphere.

2.6. Soil evolution and endemism

At least as far back as Rode (1947), pedologists were conceptualizing pedogenesis as soil evolution, vs. soil development. As the distinction is drawn by Huggett (1998), developmental approaches are based on the idea of progressive development until some sort of equilibrium with the environment is reached. Evolutionary approaches, by contrast, recognize multiple developmental pathways, the possibility of persistent nonequilibrium, and both regressive and progressive pedogenesis. Johnson and Watson-Stegner (1987) developed a model of pedogenesis along these lines, which has been implicitly or explicitly reflected in several recent conceptual models of soil and regolith formation (Schaetzl and Anderson, 2006).

The concept of soil endemism seems to have been first mentioned by Guo et al. (2003) in their study of pedodiversity in the USA. They

noted that at the lowest level of soil taxonomy, soil taxa increase continuously with area, indicating a considerable number of soils which are geographically restricted (i.e., endemic). Bockheim (2005) examined soil endemism and its relationship to pedological theory. He suggested direct parallels between neo- and paleoendemic organisms and endemic soils, as well as a number of other parallel concepts between biogeography, ecology, and evolutionary biology on one hand and pedology on the other (Bockheim, 2005). Soil evolution is listed as the pedological equivalent of biological evolution, with the former defined as "change in a soil body because of change in environmental conditions" (Bockheim, 2005: 111). This could indeed be considered analogous to selective pressure from the environment, but does not directly address the phenotypic expression of biota via soils.

2.7. Synthesis

To the extent soils are biotic constructs, and include biomantles, biological variations and changes should be reflected in soils. Similarly, the concepts of niche construction and ecosystem engineers indicate that biotic influences on soils may be both a cause and result of selective pressures. The notion of extended phenotypes further suggests that these pedological manifestations of biological effects are genetic expressions. However, because most, if not all, soils are influenced by several organisms, the term extended composite phenotype is used to reflect that few soils are likely to be extended phenotypes of any single organism.

A model of soils as extended composite phenotypes–and reflecting natural selection–also arises from biogeomorphological notions of coevolution of landforms, biota, and soil. If the biosphere is seen as a planetary membrane for capturing and transforming solar energy, then soils may also be conceptualized as a biologically excited skin or membrane, which in turn is at least potentially subject to selective effects.

Fig. 1 summarizes how the conceptual frameworks discussed above instigate a model of soils as extended composite phenotypes.

3. Paleosols and coevolution

If soils are indeed extended composite phenotypes, then there should be some evidence of fundamental changes in soils associated with biological evolution. Paleosols show different biochemical signatures associated with different vegetation (for example isotopes reflecting differences between C3 and C4 plants), and some paleosols are fossil-bearing. The extended composite phenotype concept, however, requires that beyond containing evidence of biological change, different soil types or morphologies should be associated with biological evolution. This section is not intended as a comprehensive review of paleopedological evidence of coevolution of soils and organisms, but rather to present an illustrative selection of evidence.

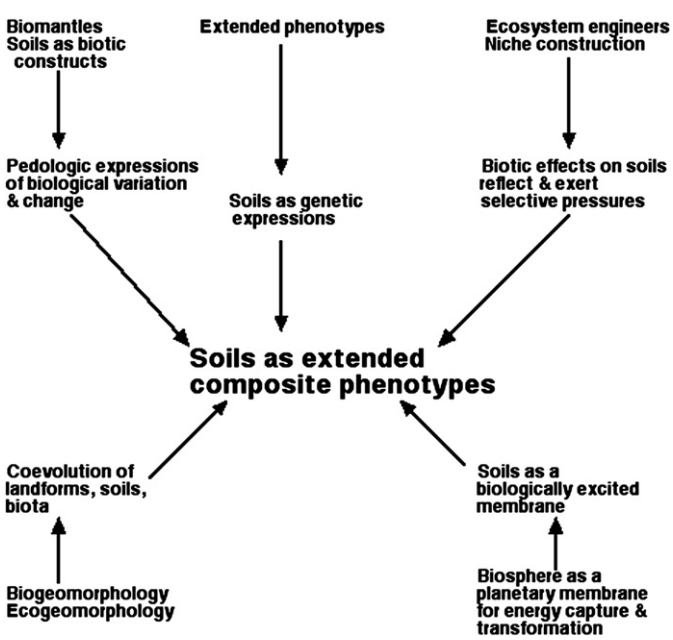

Fig. 1. Conceptual pathways converging on soils as extended composite phenotypes.

"It may be objected," Retallack (2004a: 566) wrote, "that soils are merely altered rocks and sediments, so not comparable with evolving organisms, but that is a 19th-century view effectively disproved by Dokuchaev" in 1883. Dokuchaev (1883) showed that grassland soils have specific morphological properties attributable to vegetation, organic matter, and earthworms regardless of the parent material or geological context. Subsequently, others (e.g. Retallack, 2001, 2004b) have shown that grassland soils have coevolved over the past 40 Ma with grasses and herbivores.

The term pedotype was coined by Retallack (1994) to refer to paleosols as recognized in the field, specifically noting that "each pedotype can be considered a trace fossil of a different kind of local ecosystem" (p. 1377). The establishment of grassland vegetation in North America, as reflected in the stratigraphic record, is associated with the appearance of new grassland pedotypes in the Eocene in North America. Paleosol evidence, consistent with other fossil evidence, shows emergence of soils associated with bunch grassland, sod grassland, and sagebrush as the vegetation types emerged at various times between 39 and 6 Ma, with paleosols reflecting the apparent coevolution of vegetation communities and mammalian fauna, along with abiotic factors (Retallack, 2004b, 2007).

The stratigraphic record in lower Eocene rocks of northwest Wyoming shows a direct correlation between the decline of old and appearance of new faunas (i.e., faunal turnover events) and paleosol development (Brown and Kraus, 1993), with geomorphological, pedological, and biological processes responding to same environmental forcings. This is consistent with the notion of selective pressures on biota being expressed in soils.

Retallack (1994) linked pedotypes associated with the Cretaceous–Tertiary boundary in eastern Montana with paleoenvironmental settings and associated plants and animals. In mid-Cenozoic paleosols of central Oregon, Retallack (2004a) showed how the ecological polarities (tendency toward various ecological strategies) of communities and ecosystems could be worked out for paleosols, as well as for organisms whose fossil remains occur in the paleosols.

The review of paleosols in the context of the evolution of atmospheric oxygen by Rye and Holland (1998) shows that Fe loss in soils not only reflects changes in the redox status of the atmosphere (itself linked to the evolution of a biosphere), but also more direct biological effects, resulting in systematic changes in soils paralleling the evolution of vascular plants. The proposed extension of the paleontological record of methanotrophs on land by more than 1.5 Ga by Rye and Holland (2000) is based on the pedologic signatures of methanotrophy in a 2.76 Ga paleosol in western Australia.

The Permian–Triassic mass extinction event, characterized by distinct changes in fossil assemblages across the stratigraphic boundary, is also characterized by distinctly different assemblages of paleosols (Retallack et al., 2003, 2006; Sheldon, 2006).

Quade et al. (1995) used stable isotope evidence from paleosols to reconstruct late Miocene environmental change in the Himalayan foothills. The paleosols they describe also contain basic morphological differences in the stratigraphic section which reflect vegetation shifts. Dergacheva (2003) showed that humic and fulvic acids in modern and paleosols reflect bioclimatic conditions—and associated biotic communities. Soil acidity in turn affects soil biota. She used

these properties to reconstruct Quaternary interglacial environmental changes in Siberia.

Brief as it is, this sampling of the paleopedological record does show evidence of coevolution of soils and biota. The fact that this is often reflected in pedotypes (ecosystem fossils) and community turnover is consistent with soils as extended *composite* phenotypes, rather than as expressions of any single organic/genetic influence.

4. Tree–soil interactions

Knoll's (2003) discussion of the geological consequences of biological evolution highlighted the importance of the emergence of woody plants. Wood represented a substantial innovation with respect to storage of carbon and other nutrients, and trees and shrubs had important consequences with respect to increased penetration of substrates by roots. Interactions between trees and soils were considered in an evolutionary framework by Binkley and Giardina (1998), who identified three degrees of tree–soil coupling: tightly-woven interactions, where the fitness of the tree is enhanced by its effects on soils; loosely-woven relationships, where selection pressures on trees independent of soil characteristics leads to indirect effects on soils which might positively or negatively influence fitness; and frayed connections, where tree effects on soils are unrelated to selection for tree fitness.

If soils are indeed extended composite phenotypes, this leads to the hypothesis that significant biological changes such as forest vegetation composition should be reflected by significant qualitative changes in soils (in addition to quantitative changes in soil properties). That is, selective pressure on organisms (in this case trees) should ultimately be reflected in non-transient changes in soils such as soil morphology and stratigraphy. The forests of the Ouachita Mountains, USA, offer an opportunity to test this hypothesis. A dominant selective factor, fire frequency, has been manipulated by humans to produce changes in vegetation composition from closed-canopy hardwood-dominated forests to pine-dominated woodlands. There also exists a wealth of information from recent studies on the influence of trees (both in terms of forest cover and individual trees) on soil morphology and stratigraphy, including some specific contrasts between pines and hardwoods. This information was reexamined in the context of testing the hypothesis above.

The hypothesis assessed here–that biotic changes due to selective pressures are associated with significant changes in soil type and morphology–is a necessary but not sufficient condition to support the proposal that soils are extended composite phenotypes. At present the vegetation effects on soils, and the general tree–soil feedbacks in the study area, are beginning to be understood. However, the specific selective effects of soil changes on trees and other organisms require paleopedological studies and/or incorporation of genetic analyses into studies of tree–soil interactions.

Information currently available for the study area is based on studies of modern processes and recent historical changes, and on a more general understanding of Quaternary vegetation change and soil development. From this it has been shown that effects of individual trees on soil morphology create micro- or patch-scale biotopes which favor reestablishment of trees, such that trees preferentially reoccupy the same sites, rather than a more random or uniform occupation of space within a stand over multiple generations of forest (Phillips and Marion, 2004, 2006; Phillips, 2008). Other studies in southern U.S. forests indicate that these effects would likely favor the establishment of the same species (Van Lear et al., 2000). The pedological influences of trees in the Ouachitas clearly influence important edaphic factors such as soil thickness, rock fragment content and distributions, soil horizonation, and organic matter (Phillips and Marion, 2005; Phillips et al., 2005a). Beyond this, however, the influence of soil changes on natural selection remains to be worked out.

4.1. Study area

The Ouachita Mountains (Fig. 2) are parallel, east–west trending ridges characterized by a humid subtropical climate. The soils, described in detail elsewhere (Phillips and Marion, 2004, 2005), are predominantly Hapludults. Surface textures are generally loam or sandy loam, while subsoil textures range from sandy clay loam to clay. Not unexpectedly, broad textural variations reflect the lithology of the parent material.

Data and observations reviewed here are based on sites in Arkansas, within Ouachita National Forest (ONF). Current forest vegetation in the ONF consists of oak-hickory (i.e., hardwood-dominated), shortleaf pine (pine-dominated), and oak-pine (mixed pine-hardwood) forest types. The pine-dominated sites include pine-bluestem woodlands, the dominant community on southern exposures at the time of European settlement, and sites where shortleaf pine dominates the overstory, but where hardwoods may dominate the mid- and understory.

Shortleaf pine (*Pinus echinata*) is of primary interest here. Loblolly pine (*P. taeda*) and shortleaf/loblolly hybrids are also present. The hardwoods are chiefly oaks (*Quercus* spp.), particularly southern red oak (*Q. falcata*), white oak (*Q. alba*), northern red oak (*Q. rubra*), and post oak (*Q. stellata*). Mockernut hickory (*Carya tomentosa*) and other hickories (*Carya* spp.) are also common, as is sweetgum (*Liquidambar styraciflua*).

4.2. Vegetation history

Delcourt and Delcourt (1991) synthesized paleoecological studies of the Interior Highlands region which includes the Ouachitas, covering the ~20 ka since the last glacial maximum (LGM). At the LGM spruce forests dominated the Arkansas landscape, including both uplands and lowlands. A diverse suite of other species was present, but was restricted to their ideal habitats. Oak, hickory, and a limited number of pines were relegated to warmer microclimates. Southern pines such as shortleaf and loblolly do not register in the pollen profile until later in the Holocene. Eventually, the spruce forests that typified the glacial maximum were slowly supplanted by jack pine, followed by northern pines (Delcourt and Delcourt, 1991; Wright, 1981).

About 12 ka general climate warming led to establishment of prairie in eastern Oklahoma. This easterly migrating prairie served as a phytogeographic barrier to the westward movement of eastern deciduous forest and dispersal of boreal conifers in the north (Delcourt and Delcourt, 1991). Bryant and Holloway's (1985) analysis of pollen from a site in Ouachita Mountains shows high percentages of grass (Poaceae), ragweed (Ambrosineae), and other herbs in the aster family (Asteraceae), consistent with arrival of prairie vegetation with the transition into Holocene climatic conditions. Furthermore, pollen analysis shows an abundance of grasses persisting until approximately 5 ka, diminishing fairly rapidly thereafter (Bryant and Holloway, 1985).

The decline of grasses was associated with a steady rise in southern pine, oak, and hickory. Eventually, the oak, hickory, and pine forests, once restricted to warmer sites, became more prominent (Bryant and Holloway, 1985; Delcourt and Delcourt, 1991). Shortleaf pine's ecological tolerance to a wide range of edaphic and synoptic conditions provided to be a competitive advantage over other species. This allowed shortleaf to move into southeastern Oklahoma, northern Arkansas, and southeastern Missouri during the hypsithermal interval's (up to ca. 4 ka) increasingly dry conditions. Schmidtling (2007) traced the post-Pleistocene evolution and migration of southern pines in the U.S. Based on geographic trends in adaptive traits and genetic variability, Schmidtling (2007: 31) proposed that *P. echinata* had a more or less continuous distribution across the Gulf of Mexico, extending to areas now on the continental shelf. With Holocene warming the species spread to its current range.

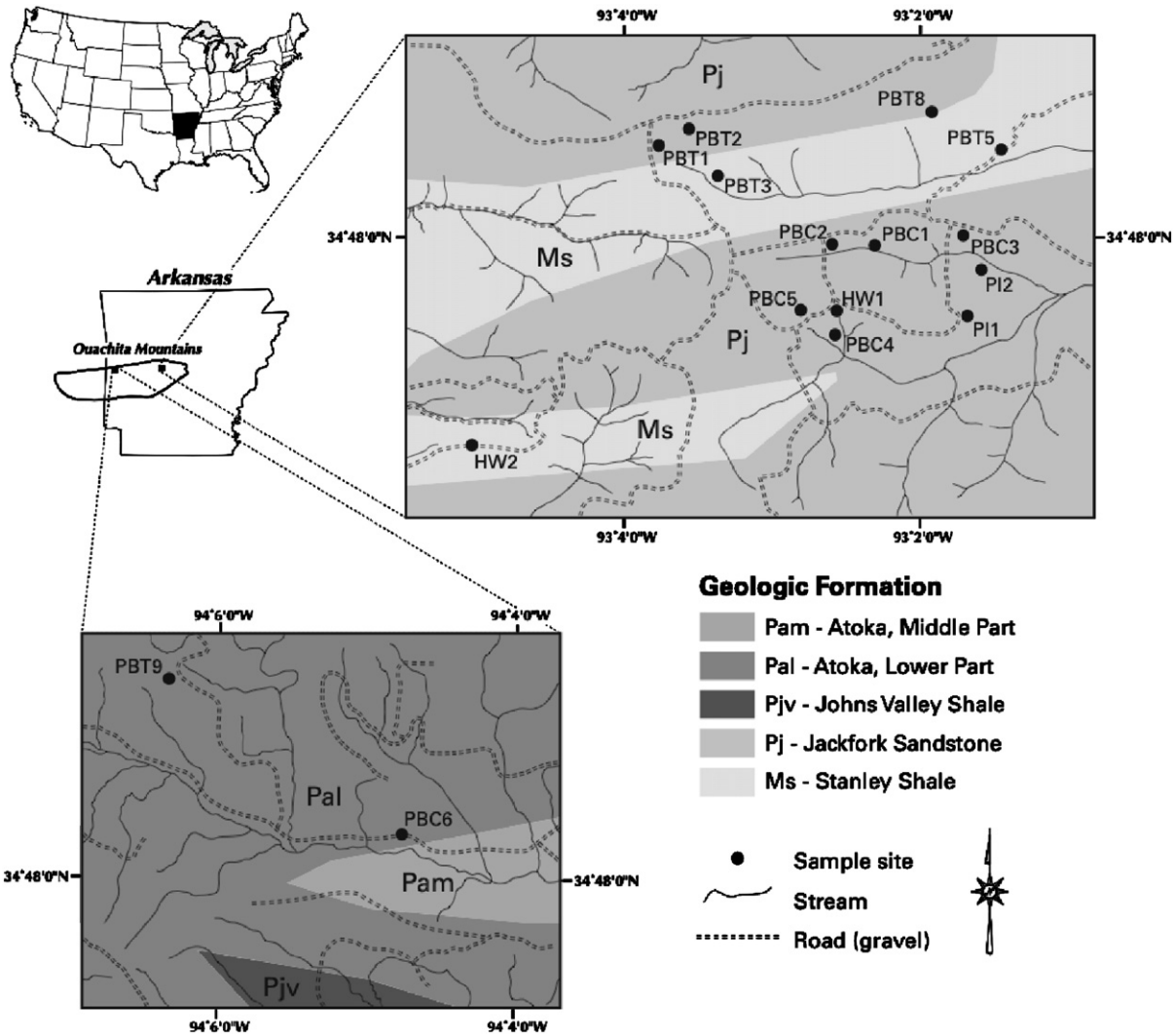

Fig. 2. Study sites in the Ouachita National Forest used to examine tree–soil interactions. Sample site designations, as well as the sites themselves, are described by Phillips and Marion (2004, 2006).

Shortleaf pine woodlands were common in the period before European settlement, largely due to a regular burning regime of frequent, low-intensity fires set by Native American people. Fire suppression following European settlement, and especially in the 20th century, led to the replacement of these woodlands with mixed pine-hardwood forests (Foti and Glenn, 1991; Davenport, 2008; Guyette et al., 2008). The selective pressure of frequent burns is discussed further below.

4.3. Fire and selective pressure

Though some of the hardwood species in the Ouachitas are fire tolerant, the fact that shorter fire intervals favor shortleaf pine in the region is well established (Guyette et al., 2008). Burning at <5 year intervals is in fact the primary method in contemporary efforts to restore pine woodlands (Davenport, 2008).

Fire may act as a selective force on plants both through its direct effects by killing or wounding susceptible individuals and through its effect on the environment: the post-fire environment may select specific physiological traits or life histories (Schwilk and Ackerly, 2001). McCune (1988) examined the ecological diversity of North American pines, classifying *P. echinata* among a group of southern pines on mesic sites exhibiting rapid growth, heavy wood, long leaves (needles), and short needle retention. He found shortleaf pine to have characteristics of fire resilience due to cone serotiny, small seeds, and precocious reproduction. *P. echinata* is classified as fire-tolerant by the U.S.D.A. Forest Service. The species survives low- and medium-intensity fires well, and regenerates well after fires (Carey, 1992). Evolutionary and phylogenetic aspects of fire-selection in pines is discussed by Schwilk and Ackerly (2001).

4.4. Pines vs. hardwoods

Recent work on the interactions among vegetation, soils, and landforms in the ONF has examined, among other things, the influence of trees on soil formation, morphology, stratigraphy, and pedodiversity, and on specific properties such as thickness (depth to bedrock), horizonation, pH, and vertical rock fragment distributions (Phillips, 2008; Phillips and Marion, 2004, 2005, 2006, 2007; Phillips et al., 2005a,b, 2008a). Some of this work relates to forest community composition and to the effects of trees in general. Of specific interest here are contrasts between pines (especially *P. echinata*) and hardwoods.

In some situations the taproot architecture typical of many pines, and coarse root architecture in general, is a key determinant of tree effects on soils. However, at the ONF study sites both pines and hardwoods exhibit taproot morphologies, and the relatively shallow soils limit the penetration of all tree roots. Thus the major contrasts

between pines and hardwoods with respect to soil morphology in the study area are:

(1) Pines are the first trees to colonize exposed rock surfaces, and to begin exploiting joints and other rock partings, initiating weathering and pedogenesis (Phillips et al., 2008a). On bedrock benches exposed by dam spillway construction about 25 years earlier, 76% of the trees at one site were shortleaf pine, which comprised 100% of trees with diameter at breast height (dbh) >5 cm. At another site, *Pinus taeda* comprised 53% of trees, with hardwoods accounting for only 8% (the remainder were red cedar, *Juniperus virginiana*, a common colonizer of disturbed sites). The shortleaf vs. loblolly pine/red cedar contrast at the two sites reflects the dominant seed sources on adjacent forested slopes.
(2) Pines are more likely than hardwoods to uproot in the Ouachitas. In other settings such as the Atlantic and Gulf coastal plains, pines are less susceptible than hardwoods to uprooting due to the deeper taproots which typically develop. In shallower Ouachita soils, however, pines are more susceptible to uprooting during ice storms due to the greater surface area for ice collection. Inventories of trees uprooted by an ice storm in 2000 showed that almost all uprootings involved shortleaf pine (Phillips et al., 2005a). Shortleaf pine is also more vulnerable to uprooting due to windthrow. Inventories at several locations where tornado blowdown occurred in mixed pine-hardwood stands in 2005 showed that all uprooted live trees were pines. Hardwoods were either left standing or suffered trunk breakage (Phillips et al., 2008b).
(3) Soil is mounded beneath shortleaf pines, while little or no mounding occurs around hardwoods. This influences both the style of soil and rock fragment displacement, and the mode of infilling of stump holes (Phillips and Marion, 2006).
(4) Shortleaf pine is faster-growing and has a shorter lifespan than the common hardwoods in the area. This results in a more rapid rate of biomechanical mixing of soils.

4.4.1. Effects on soil morphology

The effects of different tree species on soil chemistry, biology, and nutrient regimes have been examined in the context of recent efforts to restore pine-bluestem woodlands in the ONF. Vegetation change does result in significant change in these factors (Masters et al., 1993; Stanturf et al., 2000; Liechty et al., 2004, 2005). The emphasis here, however, is on longer-lasting morphological properties that result in development of qualitatively different soil types (see Phillips and Marion, 2007).

The role of pines as early colonizers of rock surfaces and thin soils (item 1 above) is important in the development of a soil/regolith cover (Phillips et al., 2008a), and the effects of individual trees are critical in soil deepening, and in determining variations in soil thickness (Phillips and Marion, 2004, 2005; Phillips et al., 2005b; Phillips, 2008). It could be expected that without pines formation of soils from weathered rock would occur more slowly, but no distinct pedological signature of pines vs. hardwoods is evident due to this difference.

One important mechanism for soil deepening by trees is via uprooting. Coarse roots typically extend into weathered bedrock and penetrate joints in unweathered bedrock (Phillips et al., 2005b, 2008a,b). Uprooting then results in the quarrying of rock fragments, distributing them within the soil (Phillips and Marion, 2005; Phillips et al., 2005b, 2008b). Rock fragments are common in soils of the Ouachitas, and tree uprooting significantly influences the vertical distribution of clasts within soil profiles independently of bedrock quarrying by preferentially "mining" fragments from the subsoil and enriching the surface layer (A-horizon) as rootwads deteriorate. This results in a characteristic pattern of stony surface layers overlying relatively stone-poor E, A2 or upper B horizons, with increasing rock fragment contents below (Phillips et al., 2005a). These variations are significant in terms of soil geomorphology, use, and taxonomy, and quite common in the study area (Phillips and Marion, 2005, 2007) and thus result in a soil landscape that represents the effects of uprooting, and of the increased propensity of pines to uproot relative to hardwoods.

Phillips and Marion (2006) found that soil displacement by trees in the Ouachitas varied between pines and hardwoods. Basal mounding beneath pines results in a vertical component of displacement not generally present beneath hardwoods. In these rocky soils, this makes clasts displaced by trunk growth around hardwoods more likely to fall into stump holes following tree mortality. Both pine and hardwood stumps and stump holes provide nutrient-rich microsites which favor future tree establishment (Phillips and Marion, 2004; Phillips, 2008). However, the greater tendency of pines to displace rock fragments away from the stump hole may give them a slight advantage relative to hardwoods in this form of pedological/ecological memory. The local deepening under individual trees associated with repeated occupation of the same site shown by Phillips (2008) does show some evidence of being more pronounced under pines, but data are insufficient at present for conclusive statements. If this is indeed the case, it would be an example of the tightly-woven feedback described by Binkley and Giardina (1998).

The key effects of trees on soil morphology in the Ouachita Mountains include their critical role in early stages of weathering and soil formation (Phillips et al., 2008a), biomechanical effects via uprooting, mass displacement, and infilling of stump holes (Phillips and Marion, 2006), and soil deepening via a combination of these (Phillips et al., 2005b; Phillips, 2008). Even if these effects were not preferentially associated with pines (relative to hardwoods), significant differences in growth rates and longevity could have significant effects on the rate and intensity of these processes.

P. echinata and *P. taeda* are both faster-growing and have shorter lifespans than the hardwoods common in the study area (Table 1). The more rapid growth of pines means that they become subject to potential uprooting (and, in recent decades and centuries, to harvesting) more rapidly. The shorter lifespan suggests shorter turnover times, indicating more rapid formation and creation of tree-related soil features under pines than hardwoods.

4.5. Extended composite phenotypes

The evidence summarized above indicates that Ouachita soils show abundant evidence of biological effects, of tight coupling between biological and pedological processes, and of an extended expression of phenotypes. There is no reason to suspect that similar trends do not occur in other forest soils and ecosystems. At a very broad scale soils certainly reflect the evolution of woody vegetation, and of gymnosperms vs. angiosperms. Holocene and historic vegetation change shows that a specific selective pressure (fire regime) results in changes in dominant trees. The vegetation change influences not just transient soil properties, but fundamental qualitative features of soils. These in

Table 1
Growth rate and lifespan categories, maximum height (hgt) after 20 years, and maximum height at 20 years as a function of typical height at maturity for common pines and hardwoods found at the ONF study sites

	Growth rate	Lifespan	Max hgt after 20 years (ft)	hgt after 20 years/ hgt at maturity
Pinus echinata	Rapid	Moderate	60	0.60
P. taeda	Rapid	Moderate	50	0.50
Quercus alba	Slow	Long	25	0.25
Q. falcata	Slow	Long	35	0.35
Q. rubra	Moderate	Long	36	0.36
Q. stellata	Slow	Long	25	0.31
Carya tomentosa	Slow	Long	18	0.18
Liquidambar styraciflua	Rapid	Long	45	0.45

Information from the U.S. Department of Agriculture Plant Database (www.plants.usda.gov).

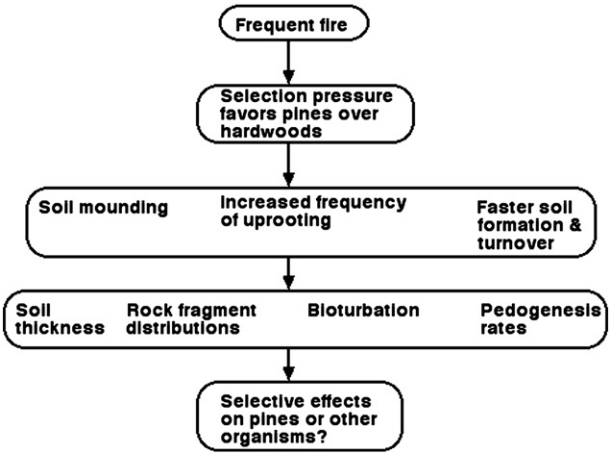

Fig. 3. Summary of how selective effects of fire influences soils in the Ouachita study sites.

turn influence vegetation, certainly in a general edaphic sense and perhaps with respect to selective pressure. This sequence is summarized in Fig. 3.

5. Discussion and conclusions

The notion of soils as biomantles and biotic constructs indicates that biological variations and changes may be manifested in soils and regoliths. The model of ecosystem engineers supports a view of soils as integrated ecosystems structured by keystone or dominant engineering species. The niche construction model includes an additional level of complexity by suggesting that pedologic influences of biota both reflect and exert selective pressures on the engineer species, and other species in the ecosystem. Pedologic responses to an organism's activities and effects may thus be extended genetic expressions. These ideas, together with those of coevolution of landforms, soils, and organisms, and those of the biosphere and soils as planetary skins or membranes which capture and transform energy, collectively point to a view of soils as reflecting the cumulative genetic effects of multiple generations of multiple organisms—soils as extended composite phenotypes. Both positive and accidental niche construction contribute to genetic expression at the scale of the ecosystem which connects the evolution (i.e., coevolution) of some, several or all the species present in the soil though the modification of physical parameters. These dynamics have important implications because physical components of soils connecting species co-adjust on the long term with organisms' evolution. This leads to a consideration of abiotic and biotic components of soils as parts of an integrated system regulated by bottom-up and top-down feedbacks from molecular scales (DNA) to the ecosystem level.

This view is not intended to supplant other ways of perceiving, defining, and analyzing soils. Rather, the intent is to propose a view of soils useful in addressing the coevolution of biotic and abiotic components of the environment. This is relevant not just to pedology and other soil sciences, and to paleopedology and paleoecology, but to the use of soils and paleosols as environmental indicators, and to geological, geographical, archaeological, and ecological interpretations of soils.

Advances in earth system science and biogeochemistry have led to an increasing appreciation of tight couplings between and coevolution of the biosphere and the atmosphere. Soils as extended composite phenotypes imply a similar relationship between the bio- and pedospheres. Further, as the physical, functional, and conceptual link between the lithosphere and the atmo-, bio-, and hydrospheres treating soils as extended composite phenotypes is consistent with studies of Earth as a tightly coupled set of interconnected systems.

The ECP framework also suggests initial possibilities with respect to searches for life on other planets. A living biosphere would likely have a signal in atmospheric chemistry, but an extinct biosphere (such as may have existed on Mars) would be unlikely to leave its fingerprints in atmospheric composition. Similarly, a biosphere's effects on soil chemistry are likely to be transient, whereas impacts on soil morphology would be more long-lasting. Thus an explicit view of soils as ECP may yield useful clues about potential extraterrestrial pedologic signatures of life.

Science is guided by its metaphors. Adoption of different metaphors may lead to new and different insights. Soil and earth sciences have often employed a notion of soil as a surficial mantle of weathered debris. Soils as ECP point away from the relatively static and inorganic mantle/blanket metaphors to more dynamic and organic metaphors such as Nikiforoff's (1959) “excited skin,” Vernadsky's (1926) planetary membrane—and, of course, Geoderma.

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